

Dear participants in the team projects of Mr. Webert and me

We will soon set a date and room for the presentation of the results of your seminar projects. All of you should be present at all presentations to get an impression of what the other groups have done.

We will afterwards go over to the lab where you shall demonstrate us the practical results.
We will additionally need a documentation and/or videos of your project.

Here is some information concerning the **presentation**, the **documentation** and the **videos**:

Please observe the following notes when creating your **presentation**:

You present as a group. However, each member of the team shall present within 5 min his or her tasks/results within the project. We will ask questions directly after the group presentation and within the practical demonstration concerning the project.

You should use PowerPoint for your presentation. Please take into account that you usually need 1-2 minutes per slide, which limits the number of slides. Focus on the important stuff. Remember that listeners are not so deep in your project as you are, so that you should start first with overviews, the basic questions of the task, possible applications, etc.

After the presentation you should save the presentation in the project output folder on the network.

1. The presentation is also intended to "pick up" listeners, for whom the topic is unknown, and to inspire them for the topic. Your own contributions and the challenges must be clearly identifiable.
2. The following points serve as guidelines for the structuring
 - Introduction, motivation / problem / tasks / conceptual formulation
 - Requirements for the project
 - Approach / concept / overview of the implemented solutions / architecture
 - Selected implementation aspects (how did you implement special points)
 - Evaluation of the results
 - Outlook
 - Literature
3. Things you should avoid:
 - Poor slide layout (slides which too full, i.e. have too much text and/or too much pictures, plain text slides, text with a font size which is too small (less than 16 pt.), defocused, unrecognizable images, too much animation, missing slide numbers, etc.)
 - Confused structure, missing common theme, not kept to the central theme
 - Missing overviews, too fast to deep in detail
 - Missing or unrecognizable presentation of your own work contribution
 - Too many slides (count at least 1-2 minutes per slide)
 - Pure reading of the slides during the presentation
4. As a training in advance to the official presentation, give your presentation at least 3 times, best before an audience, with measuring the time. If your presentation is considerably longer or shorter than 5 minutes, you will get a point deduction.
5. Test the presentation before using a beamer! Very often, you can only detect problems when using a beamer e.g. certain color combinations do not work, images are no longer recognizable or text is too small.
6. If possible, we will hire trainers from the trainer pool of the h_da, which will record your presentation to give you a feedback from their non-technical point of view after your talk.

Please observe the following notes when creating the **documentation**:

You have to create a technical documentation your project - the technical content/your approach/your realized algorithms or hardware, etc. - in form of a written documentation and a video of the results or a step-by-step-tutorial-video and a video of the result.

You should complete the documentation to the presentation date (Deviations from this, you must be discussed with us.). Here you should go more into the detail as within the presentation and show for example screenshots, program expressions, etc.

You shall hand us over the PowerPoint presentation and the documentation/video after the presentation (via USB stick) or save it in the output folder of your project on the project server.

The result of your work should be shown in a result video or within the documentation (as decided during the project together with your supervisor). This result video/document part can be finished and placed on the project output folder after the exams (date should be defined in advance together with the supervisors).

Please observe the following notes when creating the **written documentation**:

1. The report should include a table of contents.
2. It should contain a header and a footer (including page number).
3. It should begin with a description of the status of the process/line/plant/technology at the start and the objectives of the project.
4. In the report, your project should be classified in the state of the art, i.e. you should describe what your project result should serve and what the overall objectives are. You should also write down the specific goals and tasks of the project (see project presentation).
5. Perhaps you have noticed in your project that it is not always easy to restore the project result of a previous group in order to build on it. To enable future project groups to build on your work, please consider the following points:
 - a) The documentation should be written in such a way that students who work on a following seminar project based on your project have a simplified start and do not step in the same trap as you did. Your documentation should also help us to understand your work if we want to eliminate errors or if we want to add additional features. For some projects, a step-by-step statement or video (for example, working with a program) is useful. If this video is an alternative to the documentation it has to document all necessary points as a written documentation would do. Please discuss the alternatives of creating a written documentation or a video with your supervisor.
 - b) At the end of your project, a complete development environment should be installed on at least one laboratory PC, possibly distributed over several laboratory PCs. You should have created the last project status completely with that development environment. Please document on which PC/PCs (project PC numbers and project PC IP address(es)) which version of the development system is running. If the development takes place on several PCs, also explain on which PC you have developed which part of your project. Also document how you start the development environment and the development process (with a short video if necessary). Finally, write down all the necessary passwords (only in the text and not in the video). This may include how to load/deploy your project result to the target hardware.
 - c) If you have installed development tools or development environments yourself, document where you have downloaded which version of which tool, what requirements are necessary for the installation and how you have installed and configured the tools (incl. self-created or customized manifest files, etc.).

6. So that we can start your project result independently, e.g. for demonstration purposes, document how and on which target hardware in the laboratory your project result is started (if necessary with a short video), i.e. on which PC, with which link, which settings and if necessary, but only in text, with which passwords.
7. The report should outline the responsibilities of each team member for the subprojects and work packages. We also want to see in the documentation who created which part of the documentation, e.g. by displaying it in the footer of the respective section.
8. The report should end with a conclusion on the tasks completed and the status of the process/line/plant/technology at the end. Subsequently an outlook with meaningful enhancements for the future should follow. Finally follows a list of all sources and or a bibliography.
9. Technical contexts are described in a short way without embellishments. Do not use first-person narrative (not I, me, us, them ...) but use the passive form (e.g. "The description process is activated by pressing the start button ...") or an impersonal description (e.g. "This formula assumes boundary conditions ... " or " One should proceed as follows ... ").
10. Quotations: Text elements, which are copied 1:1 from an external source, have to be placed in quotes. The source has to be mentioned after the quoted block. If text elements or figures are not taken word by word, the quotes are omitted. Every quotation has to be identified by a unique identifier, consisting of a 3-5 letter abbreviation of the author and the year. We recommend the Harvard referencing rules. A quick guide to the Harvard referencing with citation examples can be found on <https://www.scribbr.co.uk/referencing/harvard-style/> or in a german version with examples with drop down menu on <https://www.scribbr.de/harvard-zitierweise/literaturverzeichnis-harvard/>.

Examples for citations: (*First author surname, second author surname and third author surname, Year*), e.g. (Doe, 2019) or (Doe and Smith, 2021)

All citation sources should be included in the reference list or in the bibliography. Use the Harvard referencing for the entries as well.

For a book use *First author surname, initial., Second author surname, initial. and third author surname, initial. (Year): Book title, City, Country: Publisher, e.g.*

Doe, J. and Smith, S. (2021): Industry 4.0, Berlin, Germany, Springer

For referencing a website use *Author surname, initial. (Year): Page Title, available at: URL (Accessed: Day Month Year)*, e.g.

Brownlee, J. (2022): Introduction to the Python Deep Learning Library TensorFlow, available at: <https://machinelearningmastery.com/introduction-python-deep-learning-library-tensorflow/> (accessed: 21 November 2022)

The reference list or the bibliography follows directly after the last chapter Conclusions, Summary or Outlook.

The information in the bibliography must be complete. Readers (supervisors or subsequent industrial processors, students if applicable) must be enabled to find the literature references on the basis of the information provided.

11. **Figures also need a source, if you did not create the figure yourself!**
12. You do not have to rewrite information from manuals, but you can refer to it with a detailed reference (see point on quotations) including a page reference.
13. **Use of generative AI:**
 - a) Documentation of the use of generative AI and/or AI-assisted technologies used during the project.
The use of generative AI and/or AI-assisted technologies for the the project is restricted for reasons of research integrity, which are based on guidelines from research and industry. In general, the following applies:

- Authors must indicate in the documentation whether and how generative AI and/or AI-assisted technologies were used during the project.
In the documentation, include a comment (either directly in the text, as a footnote, or as an extra section) for each activity (including research, programming support, etc.) in which you used generative AI or AI-assisted technologies, and/or for each result you achieved with generative AI or AI-assisted technologies, stating the following:
For [name of activity] / To achieve [name of result] [select the appropriate option from the previous list], the authors used [NAME OF TOOL/SERVICE] for [REASON]. After using this tool/service, the authors reviewed the content and edited the content as needed and take full responsibility for the result.
- Note: To ensure that you can document this later, you should make a note of where you have used generative AI and/or AI-assisted tools already during the design and implementation phases of the project.
- b) Documentation of the use of generative AI and/or AI-assisted technologies in the documentation of the project
 - The use of generative AI and/or AI-assisted technologies is only permitted to improve the readability and language of the documentation, including spelling and grammar checks, as well as reference organizers (such as Zotero, Mendeley, etc.). The authors are responsible for reviewing all changes made by such AI-assisted tools.
 - Any other use of generative AI in the writing process is PROHIBITED. This includes uploading parts of the documentation or the entire documentation to AI tools, rewriting entire paragraphs using AI, generating AI images, illustrations, artwork, or other AI-generated content for the documentation.
 - Authors need to declare, if and how generative AI and/or AI-assisted tools have been used during the writing process. The following declarations are based on a proposal by Elsevier.
 - Insert a section titled 'Declaration of generative AI and/or AI-assisted technologies in the writing process' between the conclusion and the references.
 - In case that any of such tools have been used, please complete the following statement and include it in the section described above:
During the preparation of this work the author(s) used [NAME TOOL / SERVICE] in order to [REASON]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the documentation.
 - In case that no of such tools were used, please include the following statement in the section described above:
The author(s) declare that no AI tools or services have been used for creation of this work.
 - In case that the authors have used generative AI and AI-assisted tools during the project and/or during the writing process without proper declaration, accusation of fraud are charged with severe consequences to follow.
 - In case that the authors have used generative AI and AI-assisted tools during the project and/or during the writing process without thoroughly reviewing and revising the results as necessary, and these results show serious deficiencies, this will also result in serious consequences.

Please consider the following guidelines when creating **videos** for your projects:

1. Discuss with your supervisor if you shall create one or several videos additional or as an alternative to your written documentation. If you write a documentation additional to the video you can reference parts of the video, thus, e.g. you do not have to include screenshots in your documentation, which are already shown in the video.
2. Discuss with your supervisor if you shall create a result video, showing the result of your project or/and if you create one or several tutorial videos.
3. The video will be uploaded on our YouTube channel CRAATHDA. Thus, it will be available to a worldwide auditorium.
4. We use the software **Camtasia Version 202** as a screen recording and video editing software. Please use exactly this version and not a newer version! Otherwise, we will not be able to open your Camtasia project in case we need to rework it before publishing it!
5. Video content:
 - a) If the video is an alternative to the written documentation, it must highlight all the important points that a written documentation would include and that are necessary to understand the projects, the results, and the realization as a basis for another student project or as a documentation for troubleshooting or for an extension of the functionality. Please put yourself in the viewer's shoes during the conception of the video and therefore first give an overview of the realized process/the used technology/the used toolchain, explain the process clearly structured and the individual work steps step-by-step, slowly and in detail for a better understanding.
 - b) The video should start with the technical content as soon as possible, otherwise the viewers will switch off. Therefore, do not use the h_da intro but start the video with a slide with the video title, the names of the university, team members and supervisors. Then start immediately with the technical content.
 - c) If you document the work using a software, you should state at the beginning, either in the picture, in the spoken comment, or by a faded in text, with which version of the software you worked. If the software is open source show where you can download it. If special configuration steps or software settings are required during installation, display them as well.
 - d) Please consider in advance whether you want to create a long video or several short videos (advantage long video: The viewer only has to open one video and can jump back and forth within that video. You only need the intro and the credits once. Advantage short videos: The viewer does not immediately have the feeling that he should watch e.g. 30-60 minutes of video, which would probably deter him. The video can comment in detail on the point mentioned in the title. If the video needs to be reworked, this is much faster with a short video than with a long video (e.g. shorter duration for rendering). With several videos, the titles must be selected in such a way that it can be seen that these videos belong together. If a certain sequence of the videos is planned, this must also emerge from the titles.
If the videos refer to already existing videos of other groups (e.g. succession to an already existing video), this should be indicated in the title (e.g. Tutorial 1, Tutorial 2, ...).
If you create multiple videos, it should typically not be more than 6 for the entire project. However, you can talk to your supervisor about the number of videos.
 - e) If you want to document further project points for subsequent groups as a tutorial (e.g. how to open your project or your development interface on our computer or how to install the responsible toolchain on our computers, etc.) in addition to the tutorial video(s) to be published, you can create these in a simplified form without opening and closing credits.
6. The video needs a script, which you should write before starting to record anything. Think about what you want to show in which order, which screen recordings you need for it and

what you omit (caution, if you omit e.g. for you obvious clicks, the clueless viewer may not know how to proceed). If you work with screen recordings the software interface (gui) should be set to English. Make sure that all elements displayed in the video (e.g. in screen recordings, screenshots, photos, etc.) are clearly recognizable (texts not too small, colors clearly recognizable, contrasts clearly, etc.). Also make sure that the viewer can see on what you have clicked (on which button, folder, icon, etc.) while operating a program.

After recording, you need to edit the video well. It also needs a concise, clear, fitting off-text. You should write this text down before you record it.

Make sure that the spoken comments are understandable. The volume of the sound should be set so that the commentary can be easily understood with headphones. The sound and the volume of the comment should be similar throughout the video (e.g. use the same recording system for all sections, make sure that the distance to the microphone remains the same and that you do not record any ambient noise). If you use background music, it should not interfere with the comments. Especially if you are not commenting in your native language, be sure to pronounce clearly and speak slowly.

7. The video shall start with a title, with the name of the university, with your names and the names of the supervisors. It shall end with a gratitude if necessary and with the name of the university, with your names and the names of the supervisors.
8. Please do not show IP addresses, MAC addresses or any special IDs of our components (possibly hidden in Camtasia) in the video.
9. Please document the parameters which you used for sharing, i.e. rendering of the video (e.g. by writing down all parameter or by screenshots).
10. Take a closer look at the video at the end and check that there are no dropouts in the picture or overlaps in the sound and that the comments match the respective picture.
11. Please show us the video before you hand it over to us as we might have any wishes for changes.
12. Please copy the video AND the total Camtasia project to the final output folder of your project on our project server before or after the presentation.

Of course, you can use the same images and text sections in all documentations (presentation, videos, documentation). However, take into account that a reader may still be able to clearly figure out all details of an image in a written documentation, but that a listener of the presentation might not have a chance to read the details on a slide shown with the beamer. Please concern that if you use text within images in several documentations it shall be possible to read this text in the presentation as well as in the written documentation and the video. An exception to this would be, if the text itself is not relevant in the image, but than it would be better to have the image without the text. If important details in the image become too small, you should either enlarge the image on the page, or just show a part of the image, or show the entire picture and a zoomed part of the picture where the details are large enough either in parallel or sequentially one after the other. Please also make sure that the text can be clearly recognized and that text and images are still sharp when you enlarge them.

All items, the implementation of the project, the project result, the presentation, the technical documentation and the videos are included in the evaluation of the project.

Best regards
S. Simons & H. Webert